TECHIDS

WELCOME TO TECHIDS!

TODAY'S CHALLENGE

Build a Water Wheel!

Learn. Build. Solve.



What Does a Water Wheel Do?

Pour water on it, and it spins!



Pour water → Wheel spins → It lifts!

The water's energy moves the weight!

The Engineering Idea

1 Energy Transfer

Moving water has energy. The wheel passes it along to lift the weight.

2 Wheel and Axle

A simple machine! The wheel turns the axle, and the axle winds up the string.

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Your Challenge

Build a water wheel that can lift a weight!



Team Challenge

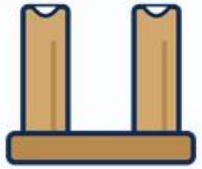
1. You have 15 minutes!
2. Work together
3. Use the provided materials
4. Heaviest lift wins!


Cardboard circles


Spoons or cups


Dowel or skewer


Straws


Cardboard frame


Tape


String


Small cup


Weights

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Time to Test!

How much weight
can your wheel lift?

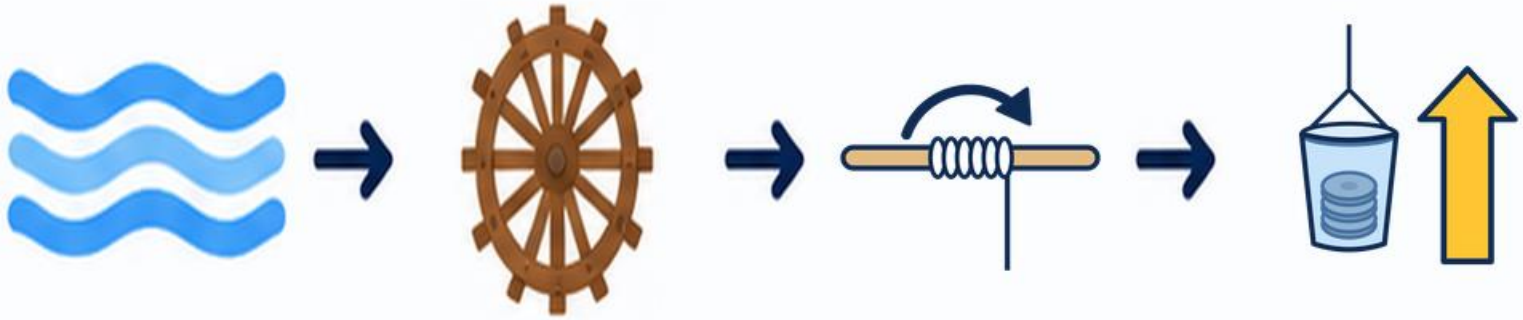
? What made it strong?



The Big Lesson

Moving water can lift a weight!

Engineers call this:
Energy Transfer + Wheel and Axle



Water pushes the paddles → Wheel spins → Axle winds the string → Cup lifts the weight